Stratified Sampling for Reducing Validation Costs of LLM-Generated Relevance Judgements*

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Abstract

Human-made relevance judgments are fundamental for the evaluation of Information Retrieval (IR) systems. However, to reduce the judgments collection costs LLMs are starting to be used for supporting or replacing humans. As a consequence, sampling-based strategies, allowing to validate LLM-generated judgements, started to be developed. Differently from existing approaches, which employ basic sampling techniques, in this work we propose using stratified sampling. Indeed, when appropriate stratification features are defined, this advanced technique allows to estimate the agreement between humans and LLMs while reducing the validation costs but still providing statistical guarantees. The experimental results show that the proposed approach allows to reduce up to 76% in the human involvement.

Keywords

Relevance Judgments, Large Language Models, Agreement Estimation

1. Introduction

Information Retrieval (IR) evaluation relies on experimental collections which feature manually annotated relevance judgments. However, constructing high-quality, realistically sized test collections is both time-consuming and expensive, primarily because of the need for manually created relevance labels. The recent advances in the development of Large Language Models (LLMs) offer the potential to reduce the cost of building IR experimental collections by assisting or replacing humans in the annotation process [2, 3, 4, 5, 6]. While the reliability of LLMs as relevance annotators remains debated [3, 7, 8, 9, 10, 11], their increasing use makes it important to develop efficient methods for assessing their agreement with human judgments. Following Merlo et al. [12], we focus on estimating the agreement between human and LLM-generated relevance judgments while minimizing the human effort required for validation and providing statistical guarantees on the estimate. The sampling-based approach proposed by Merlo et al. [12] employs Simple Random Sampling (SRS) to estimate agreement and construct Confidence Intervals (CIs) around the estimate. However, advanced sampling strategies, like Stratified Sampling (SS), can further reduce the number of human judgments required by exploiting informative stratification features [13, 14].

In this work, we identify and compare stratification features for LLM-generated relevance judgments and investigate whether SS allows to reduce human annotation costs. Specifically, we address two research questions: **RQ1 – Stratification Features**: Which features provide reliable strata for LLM-generated relevance judgments? **RQ2 – Sampling Cost**: To what extent can SS reduce the number of human judgments required to validate LLM-generated judgments?

Our results show that, with appropriate SS estimators for Mean Absolute Error (MAE), the required number of human validations can be reduced by up to 85% compared to SRS [12], while maintaining 95% confidence in the estimated agreement.

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2. Experimental Methodology

2.1. The Stratified Sampling Pipeline

Our objective is to estimate LLM–human agreement on relevance judgements while minimizing human annotations. Following Merlo et al. [12], we use an estimation pipeline based on an iterative sampling procedure but we exploit SS instead of SRS for sampling. Let $\mathscr{R}$ be the population of N LLM-generated judgments. At each iteration, one judgment is sampled and added to the sample $\mathscr{R}_{SS}$. For each sampled judgment r, we assume to have the human-made ground truth label $t(r)$. The process minimizes the number of human annotations required to estimate the agreement with a predefined Margin of Error (MoE) ϵ. In practice, we address this task through a 4-step iterative pipeline which is described in the following.

Step 0 - Stratify the Population SS relies on stratifying the population $\mathscr{R}$ into a set of H non-overlapping strata $\mathscr{P} = \{p_1, p_2, ..., p_H\}$, such that $\mathscr{P}$ represents a partition of the sample space [13]. An effective stratification allows to reduce the sample size required to achieve the desired confidence level. However, its effectiveness depends both on the allocation of individuals across strata and on the number of strata H. Indeed, a stratification is more effective the more "uniform" the individuals are in each stratum and more strata may increase homogeneity, but also require more samples. To obtain homogeneous strata we investigate five candidate features: the LLM-assigned label (A), the probability assigned to that label (P), the probability difference between the most- and least-likely labels (ΔP), the difference between the most- and second-most-likely labels (ΔP_2), and output perplexity (PPL).[1] For A, strata correspond to the assigned labels; for the continuous features, we use k-means to obtain H strata.

Step 1 - Sampling The second step of the pipeline consists of a two-phase sampling procedure. First, a stratum h is selected with probability proportional to its weight, $W_h = N_h/N$, where N_h is the size of the stratum and N is the population size. Second, a single LLM-generated relevance judgment is uniformly sampled from the selected stratum using SRS and is added to the global sample $\mathscr{R}_{SS}$.

Step 2 - Estimation of the Statistics of Interest After sampling, the point estimate and its variance must be computed using appropriate estimators which account for the employed sampling strategy. We evaluate human–LLM agreement using Mean Absolute Error (MAE), defined as the mean absolute difference between LLM-generated and human relevance labels. Since MAE is a mean, its unbiased estimator under SS is given by $\hat{\theta} = \sum_{h=1}^{H} W_h \hat{\theta}_h$, where W_h is the population weight of stratum h, and $\hat{\theta}_h = \frac{1}{n_h} \sum_{i=1}^{n_h} |r_i - t(r_i)|$ is the MAE estimated from the n_h sampled judgements in that stratum. The corresponding variance is $V(\hat{\theta}) = \sum_{h=1}^{H} W_h^2 V(\hat{\theta}_h)$, where $V(\hat{\theta}_h) = \frac{\sum_{i=1}^{n_h}(r_i - t(r_i))^2}{n_h(n_h-1)}$.

Step 3 - Constraint check The sample estimate and its variance are used to compute a CI. We employ the Wald CI [15]: $\hat{\theta} \pm z_{\alpha/2}\sqrt{V(\hat{\theta})}$, where $z_{\alpha/2}$ is the critical value of the standard normal distribution for the significance level α. Then we evaluate whether the target level of confidence in the estimate has been achieved by comparing half the width of the CI, the MoE, against a predefined threshold ϵ. If it falls below, the procedure halts; otherwise, it loops back to Step 1.

2.2. Experimental Methodology to answer RQ1

To identify the best stratification feature we fit a logistic regression model using each of the features identified above individually and we then compare the models in terms of the R^2 coefficient.[2] Notice that, while the label assigned by the LLM is a categorical value (our variable A), all the other variables

[1] Perplexity can be computed as: $PPL(X) = \exp\left(-\frac{1}{|X|}\sum_{i=1}^{|X|} \log P(x_i \mid x_1^{i-1})\right)$, where X is the LLM output and x_i is its i-th token.
[2] Being a logistic regression, we employ McFadden's pseudo R^2.

Table 1
R^2 of regression models fit using different stratification features. The symbol "⋆" indicates statistical significance at confidence level $\alpha = 0.05$ with respect to the null model, according to the likelihood ratio test.

	TREC robust 2004			TREC DL 2019			TREC DL 2020		
	Mistral 7B	Llama 8B	Qwen 30B	Mistral 7B	Llama 8B	Qwen 30B	Mistral 7B	Llama 8B	Qwen 30B
A	0.211⋆	0.622⋆	0.538⋆	0.249⋆	0.313⋆	0.310⋆	0.389⋆	0.458⋆	0.440⋆
P	0.145⋆	0.173⋆	0.256⋆	0.084⋆	0.027⋆	0.027⋆	0.147⋆	0.032⋆	0.104⋆
ΔP	0.097⋆	0.031⋆	0.000	0.023⋆	0.013⋆	0.003⋆	0.063⋆	0.010⋆	0.026⋆
ΔP_2	0.092⋆	0.113⋆	0.008⋆	0.022⋆	0.047⋆	0.017⋆	0.065⋆	0.065⋆	0.071⋆
PPL	0.040⋆	0.026⋆	0.015⋆	0.096⋆	0.017⋆	0.028⋆	0.213⋆	0.007⋆	0.043⋆
$A + P$	0.238⋆	0.634⋆	0.569⋆	0.253⋆	0.331⋆	0.311⋆	0.390⋆	0.474⋆	0.450⋆
$A + PPL$	0.225⋆	0.623⋆	0.541⋆	0.250⋆	0.313⋆	0.311⋆	0.390⋆	0.458⋆	0.440⋆

are continuous. Besides assessing individual features, we also test more complex models, combining multiple variables at once.

2.3. Experimental Methodology to answer RQ2

Inspired by Merlo et al. [12], to answer RQ2 we move to a more practical scenario. Indeed, differently from RQ1, to evaluate the estimation pipeline we do not assume to have full access to ground truth labels. We directly evaluate the full SS pipeline by measuring the annotation cost required to achieve a specified confidence level in estimating the LLM effectiveness as an annotator. Specifically, we simulate the annotation process using available experimental IR collections that include human-made relevance judgments. First we annotate all query-document pairs for relevance using an LLM. Then, we stratify the judgments using the unsupervised features described in Section 2.1. Next, we apply the sampling procedure, simulate the annotation step by using its human-provided judgment from the collection, and compute the CI. This process is repeated until the desired confidence level is achieved. Thus, a stratification strategy is considered effective if it minimizes the number of iterations needed—which directly translates into the number of required human annotations.

3. Results and Discussion

3.1. Experimental Setup

We evaluate our pipeline on the same collections used by Merlo et al. [12] but we test three LLMs: Llama 3.1 8B, Mistral v0.3 7B, and Qwen3 30B, adopting the UMBRELA prompt [3]. We set α and ϵ to 0.05, and we vary the number of strata $H \in [2, 3, 4]$. Note that, since A is categorical, when it is used alone as a stratification feature, these are the only meaningful values of H: 2, emulating binarized relevance labels [3, 16], and 3 or 4, depending on the number of relevance levels of the considered collection.

3.2. Answering RQ1: Quality of the Stratification Features

Table 1 reports the R^2 of the models fit using different stratification features to predict the correctness of the LLM labels.[3] All the models, except the one based on ΔP for Qwen 30B when applied to the TREC Robust 2004, are statistically significantly better than the null model, indicating some relation between the chosen stratification features and the response variable. If we consider single-feature models, we observe that the most expressive feature is the relevance judgement assigned by the LLM (A). In other terms, this indicates that all the LLMs tend to make mistakes more often when assigning specific classes. This aligns with past findings in the effectiveness of LLMs as evaluators [2, 17, 12]. All

[3]To favour readability, we report two configurations that include the most promising features according to the single-feature models assessment. Other approaches tend to be on a par or worse. The full table is available here: https://github.com/MerloSimone/StratifiedSamplingLLMEstimation/blob/main/complete_feature_table.pdf.

Table 2
MAE estimation results when A is used as stratification feature. The value indicates the estimated MAE, while the number between parentheses is the number of annotations required to achieve it.

	TREC robust 2004			TREC DL 2019			TREC DL 2020		
Model	**Mistral 7B**	**Llama 8B**	**Qwen 30B**	**Mistral 7B**	**Llama 8B**	**Qwen 30B**	**Mistral 7B**	**Llama 8B**	**Qwen 30B**
Real MAE	0.358	0.505	0.391	0.908	1.002	0.792	0.913	0.985	0.729
SRS	0.363 (451.4)	0.507 (1044.8)	0.391 (823.9)	0.903 (1015.9)	1.003 (1419.6)	0.785 (1151.8)	0.915 (1170.7)	0.978 (1533.8)	0.727 (1175.0)
Strata #:									
2	0.353 (335)	0.518 (272)	0.391 (302)	0.910 (895)	0.996 (1078)	0.777 (920)	0.913 (809)	0.986 (878)	0.731 (744)
3	0.360 (337)	0.521 (246)	0.388 (257)	-	-	-	-	-	-
4	-	-	-	0.920 (794)	1.006 (1008)	0.793 (855)	0.916 (639)	0.983 (748)	0.730 (642)

the other features, taken individually, appear suboptimal, with the probability assigned to the label (P) and the perplexity (PPL) being the second and third best, depending on the setting. If we move towards more complex models that include multiple features, we notice that, in most cases, there is an improvement in the fitness of the model to the data. In particular, the model that relies on the label and on the probability ($A + P$) always appears the most effective. The magnitude of the improvement over the model with only A depends on the experimental setup—despite this, it is always significant according to the likelihood ratio test that compares the two models.

The most relevant finding is that the patterns, i.e., which models tend to perform best, remain stable both across different collections and LLMs: this suggests that, if we manage to find good stratification features on historical collections, such as the ones we employed in this work, it is likely that the same features will be useful also to stratify the relevance judgements for future collections and LLMs.

3.3. Answering RQ2: Impact of Stratified Sampling on the Cost

3.3.1. MAE Estimation.

In Table 2 we report the obtained results. The first row of the table reports the real MAE value, computed by validating all the LLM-generated relevance judgments. The second row reports the estimation results and the related cost (between brackets) obtained adopting the SRS framework introduced by Merlo et al. [12]. The remaining portion reports the estimation results and costs of the proposed pipeline when different numbers of strata H is used and A is employed as the stratification feature. We omit the CI since the MoE is always equal to (or lower than) $\epsilon = 0.05$.

The results obtained reveal that, while providing a correct estimate of the MAE, the proposed estimation pipeline also allows to consistently reduce the number of relevance judgments that need to be validated by humans. In particular, for the TREC Robust 2004 collection the reduction in cost spans from 25% (for Mistral 7B) up to 76% (for Llama 8B). For the TREC Deep Learning (DL) collections, instead, the cost reduction is less pronounced and more consistent, spanning from 12% (for Mistral 7B) to 29% (for Llama 8B) for TREC DL 2019, and from 31% to 51% for TREC DL 2020. Moreover, using separate strata for each relevance level ($H = 3, 4$) allows to reduce costs, although more strata are not always beneficial. Indeed, SS sampling requires at least one judgment from each of the strata. Thus, defining more strata may lead to sampling more judgments. This occurs for Mistral 7B on TREC DL 19 where more annotations are required when $H = 4$.

4. Conclusions and Future Work

In this paper, we introduced an iterative pipeline that leverages Stratified Sampling (SS) to reduce human effort in validating LLM-generated relevance judgments. Our results demonstrate that, compared to standard sampling techniques, SS can reduce the number of human annotations required to estimate the agreement between LLMs and human assessors by up to 85%. As future work, we plan to investigate how to quantify uncertainties in IR system evaluations when relying on LLM-generated judgments.

Declaration on Generative AI

During the preparation of this work, the authors used Grammarly in order to: grammar and spelling check, paraphrase, and reword. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the publication's content.

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